

$$A_t(n) \rightarrow A_{t+1}(n) \quad t+1 \leq n$$

~~$$\forall \pi + \sigma \in S_n^{(t)} \exists \{(\pi_1, \sigma_1), (\pi_2, \sigma_2), \dots, (\pi$$~~

$$A_{n-1}(n) = A_n(n)$$

$$A_1(n) = 2n - 1$$

Number of ways to sum to $S_1^{(n)} = n-1$
 or
 $n/2$
 if order matters

What if as t and n vary
 you get the cardinality of a "shaped" set

like centered A-gen points in n dimension
 (See generated summary.mcd)